

1. Executive Summary

		Traffic Summary		Count	
		Total Flows Observed		15117	
Total Threats Mitigated		16		4691178568	
High Priority		1		4680757153	
Low Priority		15		10270	
Manual Releases		0			
Manual Blocks		0			
ML Metrics		Value		Attack Vector	
False Positives		0		ICMP Flood	
FP Rate		0.00%		SYN Flood	
IF Precision		100.00%		UDP Flood	
IF Recall		93.04%		Uncertain	
IF F1-Score		96.40%			
IF Accuracy		97.61%			
				Action	
				Quarantined	
				Time Ban	
				Blackhole	
				Released	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.04%	Share of actual attacks successfully flagged
F1-Score	96.40%	Balanced measure combining Precision and Recall
Accuracy	97.61%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.96%	Actual attacks that went undetected
True Positive Rate (TPR)	93.04%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4815	FN 360
Actual: Normal	FP 0	TN 9910

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4815	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	93.04%	Share of actual attacks successfully flagged
F1-Score	96.40%	Balanced measure combining Precision and Recall
Accuracy	97.61%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	6.96%	Actual attacks that went undetected
True Positive Rate (TPR)	93.04%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4815	FN 360
Actual: Normal	FP 0	TN 9910

2d. Response Latency

Metric	Value	Description
Detection Time	887.1 ms	Average time taken to detect an attack
Mitigation Response Time	4.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.11%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.91%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	9.05%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	2	2	2026-09-16 12:02:49	2026-09-16 12:04:00
10.0.0.17	2	2	2026-09-16 12:02:58	2026-09-16 12:04:10
10.0.0.18	2	2	2026-09-16 12:02:50	2026-09-16 12:04:01
10.0.0.19	2	2	2026-09-16 12:03:01	2026-09-16 12:04:12

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-16 12:02:49	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:02:50	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:02:58	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:03:00	10.0.0.16	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-16 12:03:01	10.0.0.18	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-16 12:03:01	10.0.0.18	DDoS	SYN Flood	93.0%	Critical	Time Ban
2026-09-16 12:03:01	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:03:03	10.0.0.16	DDoS	SYN Flood	92.6%	Critical	Time Ban
2026-09-16 12:03:09	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-16 12:03:10	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-16 12:03:12	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban (1m)
2026-09-16 12:03:12	10.0.0.19	DDoS	SYN Flood	92.2%	Critical	Time Ban
2026-09-16 12:04:00	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-16 12:04:01	10.0.0.18	DDoS	SYN Flood	93.0%	Critical	Time Ban (2m)
2026-09-16 12:04:10	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban (2m)
2026-09-16 12:04:12	10.0.0.19	DDoS	SYN Flood	92.2%	High	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6091	92.2%	High	Phase 2	2	1m 11s	2026-09-16 12:04:12	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6345	92.5%	Critical	Phase 2	2	1m 12s	2026-09-16 12:04:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6301	93.0%	Critical	Phase 2	2	1m 11s	2026-09-16 12:04:01	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6279	93.1%	Critical	Phase 2	2	1m 11s	2026-09-16 12:04:00	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6511	92.2%	Critical	Phase 2	1	11s	2026-09-16 12:03:12	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6260	92.5%	Critical	Phase 2	1	11s	2026-09-16 12:03:09	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6490	93.0%	Critical	Phase 2	1	11s	2026-09-16 12:03:01	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6383	92.5%	Critical	Phase 2	1	11s	2026-09-16 12:03:00	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE